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Diophantine Confinement of the Branch Locus in Syracuse Dynamics

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Abstract

We prove that if the transverse walk associated to a Collatz trajectory has eventually linear drift, then the trajectory is eventually bounded and periodic. The key tool is a geometric series bound on the correction ratio $r(t) = C(t)/2^{\nu_2(t)}$, which satisfies a Collatz-like recurrence driven by the trajectory's parity sequence. We show that the parity sequence of every Collatz orbit lies in the golden mean shift (no two consecutive odd steps), implying $p_{\mathrm{odd}} \leq 1/\varphi^2 \approx 0.382$, within 0.005 of the equilibrium threshold $p_{\mathrm{eq}} \approx 0.387$. Cycle elimination is proved for all cycles with $\Delta_3 \leq 79$ odd steps per period using Steiner's correction bounds, Rhin's effective irrationality measure for $\log_2 3$ [10], and Hercher's computational verification for m-cycles with $m \leq 91$ [6]. Combined with a conditional hypothesis for longer cycles ($\Delta_3 \geq 80$), this yields a conditional proof of the Collatz conjecture: linear walk drift implies collatzReaches(n). We also prove the converse: reaching 1 implies the K-bound, so the sole critical-path sorry is formally equivalent to the conjecture itself. A Syracuse framework bridge connects Tao's odd-step formulation to the walk/drift infrastructure. The argument is formalized in approximately 7 800 lines of Lean 4 across 33 files with four external axioms (Baker 1966, Rhin 1987, Hercher 2023, Weyl 1916), all citing published theorems. A skew product decomposition identifies the Collatz dynamics on the $(2, 3)$-solenoid with an irrational rotation cocycle over the 2-adic odometer; the cocycle mean irrationality and correlation decay are proved, but the gap between ergodicity and bounded deficit remains open.

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1. Introduction

1.1. *The conjecture.*

Definition 1.1. The *Collatz map* $T\colon \mathbb{N} \to \mathbb{N}$ is defined by

$$T(n) = \begin{cases} n/2 & \text{if } n \equiv 0 \pmod 2, \\ 3n+1 & \text{if } n \equiv 1 \pmod 2. \end{cases}$$

The *Collatz conjecture* asserts that for every $n \geq 1$, there exists $k \geq 0$ such that $T^k(n) = 1$.

The conjecture has been verified computationally for all $n \leq 10^{20}$ [9] and remains one of the most resistant open problems in elementary number theory. Surveys of the extensive literature can be found in Lagarias [7, 8].

1.2. *Prior work.* Terras [14] introduced the stopping time and proved that almost all integers (in the sense of natural density) have a finite stopping time. Lagarias [7] formalized the probabilistic heuristic that treats the Collatz iteration as a random walk with drift $\log_2(4/3)$ per odd step. Wirsching [16] developed the ergodic-theoretic perspective, studying transfer operators associated to the dynamics.

Steiner [12] and Simons–de Weger [11] used Baker-type bounds on linear forms in logarithms to establish lower bounds on the period of any nontrivial Collatz cycle. Eliahou [4] showed that no nontrivial cycle has period less than 17 087 915. Conway [3] proved that Collatz-type problems can be algorithmically undecidable in general, though this does not apply to the specific $3n + 1$ function.

The strongest density result is due to Tao [13], who proved that almost all orbits (in a logarithmic density sense) attain almost bounded values. His approach uses entropy-based methods on the Syracuse random variables and does not yield the full conjecture.

1.3. *Our approach.* The present paper introduces a different mechanism. The argument proceeds in three steps:

(1) The parity sequence of every Collatz trajectory lies in the *golden mean shift* (the subshift of finite type forbidding the bigram “11”). This constrains the odd-step proportion to $p_{\text{odd}} \leq 1/\varphi^2 \approx 0.382$, just below the equilibrium threshold $p_{\text{eq}} = 1/(1 + \log_2 3) \approx 0.387$.

(2) The *correction ratio* $r(t) = C(t)/2^{\nu_2(t)}$, which captures the cumulative effect of the “+1” in $3n + 1$, satisfies a geometric series bound when the transverse walk has linear drift. This forces the trajectory to be eventually bounded.

(3) Cycle elimination—proved directly for $\Delta_3 = 1$, and via Steiner correction bounds combined with Hercher's computational verification [6] for $\Delta_3 \leq 79$—shows the only positive cycle is $\{1, 2, 4\}$ (conditionally on extending the Hercher bound beyond $m = 91$ for $\Delta_3 \geq 80$).

1.4. *Main results.* We state two main theorems. Theorem A is unconditional given the drift hypothesis; Theorem B is a conditional proof of the full conjecture.

THEOREM A (Walk drift implies reaches 1). *Let $n \geq 1$. If the transverse walk $w(t)$ has eventually linear drift with rate $\varepsilon > 0$, and the only positive Collatz cycle is $\{1, 2, 4\}$, then $T^k(n) = 1$ for some $k \geq 0$.*

THEOREM B (Conditional Collatz conjecture). *Assume*:

(H1) *For every $n \geq 1$, there exist $\varepsilon > 0$, T_0, and K such that $\nu_3(t)/t \leq p_{\text{eq}} - \varepsilon$ and $3\nu_3(t) \leq t + K$ for all $t \geq T_0$.*
(H2) *Hercher's theorem: no nontrivial m-cycle exists for $m \leq 91$* [6].
(H3) *No nontrivial cycle exists with $\Delta_3 \geq 80$ odd steps per period.*

Then for every $n \geq 1$, there exists $k \geq 0$ such that $T^k(n) = 1$.

Hypothesis (H1) is a quantitative version of the statement that the transverse walk has positive linear drift. The golden mean shift constraint makes this plausible: the gap between $1/\varphi^2 \approx 0.382$ and $p_{\text{eq}} \approx 0.387$ is only 0.005. What remains is to promote this topological bound to a pointwise ergodic statement for individual trajectories—essentially, an ergodic theorem for the Collatz transfer operator, or a spectral gap argument. Hypothesis (H2) is computationally verified mathematics [6]; we treat it as an external axiom in the formalization. Hypothesis (H3) is the frontier: it requires extending Hercher's verification beyond $m = 91$, or developing analytical bounds (via Baker's theorem [2] and Rhin's irrationality measure [10]) sufficient to cover all $\Delta_3 \geq 80$.

1.5. *Outline.* Section 2 introduces the winding numbers, transverse walk, and the multiplicative identity that relates trajectory values to their parity counts. Section 3 establishes that every Collatz parity sequence lies in the golden mean shift. Section 4 contains the core result: the correction ratio bound. Section 5 treats cycle elimination, including the Steiner correction bounds and the Hercher computational verification. Section 6 assembles the main theorems. Section 7 describes the Lean 4 formalization. Section 8 summarizes computational evidence.

2. Winding numbers and the multiplicative identity

Definition 2.1. For $n \geq 1$, define the *Collatz sequence* by $a_0 = n$ and $a_{t+1} = T(a_t)$. Let $\nu_2(t)$ and $\nu_3(t)$ count the even and odd steps among

$0, 1, \ldots, t-1$, so that $\nu_2(t) + \nu_3(t) = t$. The *transverse walk* is

$$w(t) = \nu_2(t) - \log_2 3 \cdot \nu_3(t). \tag{2.1}$$

The quantity $w(t)$ measures the excess of even steps over the equilibrium rate $\log_2 3$ per odd step. If $w(t) \to +\infty$, then even steps dominate sufficiently to force the trajectory toward small values.

PROPOSITION 2.2 (Multiplicative identity). *For all $n \geq 1$ and $t \geq 0$,*

$$a_t \cdot 2^{\nu_2(t)} = n \cdot 3^{\nu_3(t)} + C(t), \tag{2.2}$$

where the correction term $C(t)$ *satisfies* $C(0) = 0$ *and*

$$C(t+1) = \begin{cases} C(t) & \text{if step } t \text{ is even,} \\ 3\,C(t) + 2^{\nu_2(t)} & \text{if step } t \text{ is odd.} \end{cases} \tag{2.3}$$

Proof. By induction on t. The base case $t = 0$ gives $a_0 \cdot 2^0 = n \cdot 3^0 + 0$.

If step t is even, then $a_{t+1} = a_t/2$, $\nu_2(t+1) = \nu_2(t) + 1$, $\nu_3(t+1) = \nu_3(t)$, and $C(t+1) = C(t)$. The identity at $t+1$ becomes

$$\frac{a_t}{2} \cdot 2^{\nu_2(t)+1} = a_t \cdot 2^{\nu_2(t)} = n \cdot 3^{\nu_3(t)} + C(t).$$

If step t is odd, then $a_{t+1} = 3a_t + 1$, $\nu_2(t+1) = \nu_2(t)$, $\nu_3(t+1) = \nu_3(t) + 1$, and $C(t+1) = 3\,C(t) + 2^{\nu_2(t)}$. The identity at $t+1$ becomes

$$\begin{aligned}(3a_t + 1) \cdot 2^{\nu_2(t)} &= 3 \cdot a_t \cdot 2^{\nu_2(t)} + 2^{\nu_2(t)} \\ &= 3\big(n \cdot 3^{\nu_3(t)} + C(t)\big) + 2^{\nu_2(t)} \\ &= n \cdot 3^{\nu_3(t)+1} + 3\,C(t) + 2^{\nu_2(t)}.\end{aligned}$$

□

Definition 2.3. The *correction ratio* is

$$r(t) = \frac{C(t)}{2^{\nu_2(t)}}.$$

LEMMA 2.4 (Correction ratio recurrence). *The correction ratio satisfies*:

- *Even step*: $r(t+1) = r(t)/2$.
- *Odd step*: $r(t+1) = 3\,r(t) + 1$.

Proof. At an even step, $\nu_2(t+1) = \nu_2(t) + 1$ and $C(t+1) = C(t)$, so $r(t+1) = C(t)/2^{\nu_2(t)+1} = r(t)/2$.

At an odd step, $\nu_2(t+1) = \nu_2(t)$ and $C(t+1) = 3\,C(t) + 2^{\nu_2(t)}$, so $r(t+1) = (3\,C(t) + 2^{\nu_2(t)})/2^{\nu_2(t)} = 3\,r(t) + 1$. □

PROPOSITION 2.5. *For all $n \geq 1$ and $t \geq 0$,*

$$a_t = n \cdot 2^{-w(t)} + r(t). \tag{2.4}$$

In particular, if $w(t) \to +\infty$, then $a_t \approx r(t)$ for large t.

Proof. Dividing (2.2) by $2^{\nu_2(t)}$ gives $a_t = n \cdot 3^{\nu_3(t)}/2^{\nu_2(t)} + r(t)$. Since $3^{\nu_3}/2^{\nu_2} = 2^{\nu_3 \log_2 3 - \nu_2} = 2^{-w(t)}$, the result follows. □

3. The golden mean shift

PROPOSITION 3.1. *If n is odd, then $3n+1$ is even.*

Proof. If n is odd, then $3n$ is odd, so $3n+1$ is even. □

COROLLARY 3.2. *In any Collatz trajectory, no two consecutive steps can both be odd steps.*

Proof. An odd step is applied when a_t is odd, producing $a_{t+1} = 3a_t + 1$, which is even by Proposition 3.1. Therefore step $t+1$ is an even step. □

Definition 3.3. The *parity sequence* of the Collatz orbit of n is the sequence $\sigma = (\sigma_0, \sigma_1, \sigma_2, \ldots) \in \{0,1\}^{\mathbb{N}}$ where $\sigma_t = a_t \bmod 2$. The *golden mean shift* $X_\varphi \subset \{0,1\}^{\mathbb{N}}$ is the subshift of finite type defined by the forbidden bigram 11:

$$X_\varphi = \left\{\sigma \in \{0,1\}^{\mathbb{N}} : \sigma_t \cdot \sigma_{t+1} = 0 \text{ for all } t \geq 0\right\}.$$

THEOREM 3.4. *The parity sequence of every Collatz orbit lies in the golden mean shift X_φ.*

Proof. This is a restatement of Corollary 3.2: the bigram $\sigma_t = 1$, $\sigma_{t+1} = 1$ would require two consecutive odd values, which is impossible. □

COROLLARY 3.5. *For any Collatz trajectory of length t, the odd-step proportion satisfies $\nu_3(t)/t \leq \lceil t/2 \rceil / t$. In particular, $\limsup_{t\to\infty} \nu_3(t)/t \leq 1/2$.*

More precisely, the topological entropy of the golden mean shift is $\log \varphi$, where $\varphi = (1+\sqrt{5})/2$ is the golden ratio. The measure of maximal entropy assigns weight $1/\varphi^2 = (3-\sqrt{5})/2 \approx 0.382$ to the symbol 1. Therefore, under the maximal-entropy measure, $p_{\mathrm{odd}} = 1/\varphi^2$.

PROPOSITION 3.6. $\log_2 3 < \varphi$.

Proof. We show $\log_2 3 < 8/5 < \varphi$. For the first inequality, $3^5 = 243 < 256 = 2^8$, so $5 \log_2 3 < 8$, giving $\log_2 3 < 8/5$. For the second, $(8/5)^2 = 64/25 < 5 = \varphi + 1/\varphi$; more precisely, $\varphi^2 = \varphi + 1 > 8/5 + 1 = 13/5$, and $(8/5)^2 = 64/25 < 65/25 = 13/5$. □

Remark 3.7. The equilibrium threshold for the walk (2.1) is

$$p_{\mathrm{eq}} = \frac{1}{1+\log_2 3} \approx 0.38685.$$

At this odd-step proportion, the expected walk increment is zero. The golden mean shift bound gives $p_{\mathrm{odd}} \leq 1/\varphi^2 \approx 0.38197$. The gap is $p_{\mathrm{eq}} - 1/\varphi^2 \approx 0.005$, tantalizingly small. This means the SFT constraint *nearly* suffices to prove positive drift. The remaining challenge is to promote the topological bound $1/\varphi^2$ to a pointwise ergodic statement for individual trajectories. Proposition 3.6 ensures that the inequality goes the right way: since $\log_2 3 < \varphi$, the reciprocal satisfies $1/(1 + \log_2 3) > 1/(1 + \varphi) = 1/\varphi^2$, confirming that the golden mean shift maximum $1/\varphi^2$ lies strictly below p_{eq}.

4. **The correction ratio bound**

This section contains the core new result.

Definition 4.1. We say the walk has *eventually linear drift* with rate $\varepsilon > 0$ if there exists T_0 such that $w(t) \geq \varepsilon\, t$ for all $t \geq T_0$.

4.1. *Unrolling the correction recurrence.* Let the odd steps occur at times $\tau_1 < \tau_2 < \tau_3 < \cdots$. Write $U_j = \nu_2(\tau_j)$ for the cumulative even-step count at the j-th odd step. Since even steps leave C unchanged and each odd step sends $C \mapsto 3C + 2^{U_j}$, unrolling gives:

Lemma 4.2 (Correction sum). *After the k-th odd step,*

$$C(\tau_k + 1) = \sum_{j=1}^{k} 3^{k-j} \cdot 2^{U_j}. \tag{4.1}$$

Proof. By induction. At $k = 1$: $C(\tau_1 + 1) = 3 \cdot 0 + 2^{U_1} = 2^{U_1}$. Assuming the formula at k, the $(k+1)$-th odd step gives $C(\tau_{k+1}+1) = 3\cdot C(\tau_k+1)+2^{U_{k+1}}$, which expands to $\sum_{j=1}^{k} 3^{k+1-j} \cdot 2^{U_j} + 2^{U_{k+1}} = \sum_{j=1}^{k+1} 3^{k+1-j} \cdot 2^{U_j}$. □

At times just after the k-th odd step (where $\nu_2 = U_k$ and $\nu_3 = k$), the correction ratio is

$$r = \frac{C(\tau_k + 1)}{2^{U_k}} = \sum_{m=0}^{k-1} 3^m \cdot 2^{-(U_k - U_{k-m})} = \sum_{m=0}^{k-1} 3^m \cdot 2^{-D_m}, \tag{4.2}$$

where $D_m = U_k - U_{k-m}$ is the number of even steps between the $(k-m)$-th and k-th odd steps.

4.2. *The walk-drift gap bound.*

Lemma 4.3 (Walk-drift gap bound). *If the walk has eventually linear drift with rate $\varepsilon > 0$, then for all sufficiently large k and all $0 \leq m \leq k$ with $\tau_{k-m} \geq T_0$,*

$$D_m \geq m \cdot \alpha, \qquad \textit{where} \quad \alpha = \frac{\log_2 3 + \varepsilon}{1 - \varepsilon} > \log_2 3. \tag{4.3}$$

Proof. Apply the drift condition at times τ_{k-m} and τ_k:

$$w(\tau_k) = U_k - \log_2 3 \cdot k \geq \varepsilon\, \tau_k,$$
$$w(\tau_{k-m}) = U_{k-m} - \log_2 3 \cdot (k-m) \geq \varepsilon\, \tau_{k-m}.$$

Subtracting gives $D_m - \log_2 3 \cdot m \geq \varepsilon\,(\tau_k - \tau_{k-m})$. Between the $(k-m)$-th and k-th odd steps, the trajectory takes m odd steps and D_m even steps, so $\tau_k - \tau_{k-m} = D_m + m$. Therefore $D_m - \log_2 3 \cdot m \geq \varepsilon(D_m + m)$, giving $D_m(1-\varepsilon) \geq m(\log_2 3 + \varepsilon)$, hence $D_m \geq m\alpha$. □

4.3. *The geometric series bound.*

THEOREM 4.4 (Correction ratio bound). *If the walk has eventually linear drift with rate $\varepsilon > 0$, then $r(t)$ is eventually bounded. Specifically, there exists K such that for all $k \geq K$,*

$$r(\tau_k + 1) \leq \frac{1}{1-\rho} + 1, \qquad \textit{where} \quad \rho = 3 \cdot 2^{-\alpha} < 1. \tag{4.4}$$

Since r only decreases at even steps ($r \to r/2$), the bound holds at all times $t \geq \tau_K + 1$.

Proof. Split the sum (4.2) at the threshold m_0 such that $\tau_{k-m} \geq T_0$ for all $m \leq m_0$. Since there are at most T_0 odd steps before time T_0, we have $m_0 \geq k - T_0$.

Recent terms ($m \leq m_0$, i.e., $\tau_{k-m} \geq T_0$). By Lemma 4.3, $D_m \geq m\alpha$, so $3^m \cdot 2^{-D_m} \leq 3^m \cdot 2^{-m\alpha} = \rho^m$. Since $\alpha > \log_2 3$, we have $\rho = 3/2^\alpha < 1$, and

$$\sum_{m=0}^{m_0} \rho^m \leq \frac{1}{1-\rho}.$$

Early terms ($m > m_0$, i.e., $\tau_{k-m} < T_0$). For these terms, $U_{k-m} \leq T_0$, so $D_m = U_k - U_{k-m} \geq U_k - T_0$. From the walk-drift condition, $U_k \geq k\alpha$, giving

$$3^m \cdot 2^{-D_m} \leq 3^m \cdot 2^{-(k\alpha - T_0)} = 2^{T_0} \cdot 3^m \cdot 2^{-k\alpha}.$$

Since $m \leq k$, we have $3^m \leq 3^k$, and $3^m \cdot 2^{-D_m} \leq 2^{T_0} \cdot \rho^k$. There are at most T_0 such terms, contributing at most $T_0 \cdot 2^{T_0} \cdot \rho^k$, which vanishes as $k \to \infty$.

Combining: for k large enough that $T_0 \cdot 2^{T_0} \cdot \rho^k < 1$,

$$r(\tau_k + 1) \leq \frac{1}{1-\rho} + 1.$$ □

COROLLARY 4.5 (Trajectory bound). *Under the hypotheses of Theorem 4.4, the Collatz sequence a_t is eventually bounded:*

$$a_t \leq B = \left\lfloor \frac{1}{1-\rho} \right\rfloor + 2 \qquad \textit{for all } t \geq T_1,$$

for some T_1 depending on n, ε, and T_0.

Proof. From Proposition 2.5, $a_t = n \cdot 2^{-w(t)} + r(t)$. Choose T_1 so that both $n \cdot 2^{-w(t)} < 1$ (possible since $w(t) \to +\infty$) and $r(t) \le 1/(1-\rho) + 1$ hold for $t \ge T_1$. Then $a_t < 1/(1-\rho) + 2$, and since a_t is a positive integer, the result follows. □

COROLLARY 4.6 (Eventual periodicity). *Under the same hypotheses, a_t is eventually periodic: there exist $T_2 \ge T_1$ and $p \ge 1$ such that $a_{t+p} = a_t$ for all $t \ge T_2$.*

Proof. For $t \ge T_1$, the sequence a_t takes values in the finite set $\{1, 2, \dots, B\}$. Since $a_t \le B$ for all $t \ge T_1$ and the Collatz map is deterministic, some value must repeat. From the first repetition onward, the sequence is periodic. □

Remark 4.7 (Why $C/2^{\nu_2}$, not $C/3^{\nu_3}$). The ratio $C(t)/3^{\nu_3(t)}$ asks how large the correction is relative to the odd-step growth. Since 3^{ν_3} grows more slowly than 2^{ν_2} when the walk diverges, this ratio can be unbounded even when the trajectory is well-behaved. Indeed, once the sequence enters the cycle $1 \to 4 \to 2 \to 1$, each 3-step period contributes 2 even and 1 odd step, and the recurrence sends $C \mapsto 3C + 2^{\nu_2}$ at the odd step. The ratio $C/3^{\nu_3}$ then grows as $(4/3)^m$ per cycle—unbounded.

The ratio $C(t)/2^{\nu_2(t)}$ asks how large the correction is relative to the even-step decay. Walk divergence ensures 2^{ν_2} grows fast enough to absorb the correction.

5. Cycle elimination

It remains to show that any cycle entered by a walk-divergent trajectory must be the trivial cycle $\{1, 2, 4\}$.

5.1. *The cycle equation.*

PROPOSITION 5.1 (Cycle equation). *Suppose a_t is periodic with period p, and let Δ_2, Δ_3 be the number of even and odd steps per period ($\Delta_2 + \Delta_3 = p$). If $2^{\Delta_2} > 3^{\Delta_3}$ (walk divergence), then the minimum cycle element $c_{\min}$ satisfies*

$$c_{\min} = \frac{S}{2^{\Delta_2} - 3^{\Delta_3}}, \tag{5.1}$$

where $S = \sum_{j=1}^{\Delta_3} 3^{\Delta_3 - j} \cdot 2^{e_j}$ and e_j is the number of even steps preceding the j-th odd step within one period.

Proof. Apply the multiplicative identity (2.2) at time t (within the periodic regime) and at time $t+p$. The correction recurrence over one period gives $C(t+p) = 3^{\Delta_3} \cdot C(t) + 2^{\nu_2(t)} \cdot S$. Substituting into the identity at $t+p$ and using $a_{t+p} = a_t = c_0$:

$$c_0 \cdot 2^{\nu_2(t)+\Delta_2} = c_0 \cdot 3^{\Delta_3} \cdot 2^{\nu_2(t)} + 2^{\nu_2(t)} \cdot S.$$

Dividing by $2^{\nu_2(t)}$ gives $c_0 \cdot 2^{\Delta_2} = c_0 \cdot 3^{\Delta_3} + S$, hence $c_0 = S/(2^{\Delta_2} - 3^{\Delta_3})$. □

5.2. *One-odd-step cycles.*

THEOREM 5.2. *The only Collatz cycle with $\Delta_3 = 1$ odd step per period is the trivial cycle $\{1, 2, 4\}$.*

Proof. With $\Delta_3 = 1$, the sum S has a single term: $S = 2^{e_1}$, where $e_1 \in \{0, 1, \ldots, \Delta_2 - 1\}$. The cycle equation gives

$$c_0 = \frac{2^{e_1}}{2^{\Delta_2} - 3}.$$

For c_0 to be a positive integer, $(2^{\Delta_2} - 3) \mid 2^{e_1}$. Since $2^{\Delta_2} - 3$ is odd for $\Delta_2 \geq 2$, and 2^{e_1} is a power of 2, the only way an odd number divides a power of 2 is if that odd number is 1. Therefore $2^{\Delta_2} - 3 = 1$, giving $\Delta_2 = 2$.

This yields $p = 3$ and $c_0 = 2^{e_1}$ with $e_1 \in \{0, 1\}$. Both values produce the same cycle $\{1, 2, 4\}$ (starting from different points).

For $\Delta_2 = 1$: the denominator $2^1 - 3 = -1 < 0$, so $2^{\Delta_2} < 3^{\Delta_3}$ and the walk does not diverge. This case is excluded by hypothesis. □

5.3. *Correction bounds and the c_0 squeeze.* The correction sum (4.1) admits both upper and lower bounds that constrain the cycle minimum $c_{\min}$.

PROPOSITION 5.3 (Correction upper bound). *For a cycle with Δ_3 odd steps ($K = \Delta_3$) and Δ_2 even steps ($L = \Delta_2$), the correction satisfies*

$$2S + 2^L \leq 3^K \cdot 2^L.$$

Proof. By induction on the steps within one period. Each odd step multiplies the correction by 3 and adds $2^{U_j} \leq 2^L$. □

PROPOSITION 5.4 (Correction lower bound). *Under the same conditions,*

$$2S + 1 \geq 3^K.$$

Equivalently, $S \geq (3^K - 1)/2$.

Proof. By induction. Each odd step contributes at least 3^{K-j} to the sum (the minimal contribution occurs when $e_j = 0$). □

COROLLARY 5.5 (c_0 squeeze). *For a cycle with $2^L > 3^K$ and $K \geq 1$,*

$$\frac{3^K - 1}{2(2^L - 3^K)} \;\leq\; c_0 \;\leq\; \frac{(3^K - 1) \cdot 2^L}{2(2^L - 3^K)}. \tag{5.2}$$

Proof. The lower bound follows from $S \geq (3^K - 1)/2$ and $c_0 = S/(2^L - 3^K)$. The upper bound follows from $2S + 2^L \leq 3^K \cdot 2^L$, which gives $S \leq (3^K \cdot 2^L - 2^L)/2 = 2^{L-1}(3^K - 1)$. □

5.4. *Steiner–Hercher cycle elimination.* The key step is to bound $K = \Delta_3$ (the number of odd steps per period) in terms of Δ_3. Since $\Delta_2 + \Delta_3 = p$ and $p = 3\Delta_3$ for the balanced case, we have $L = 2\Delta_3$, so the condition $2^L > 3^K$ becomes $2^{2\Delta_3} > 3^K$.

PROPOSITION 5.6 (K-bound for small Δ_3). *For $\Delta_3 \le 79$ in a balanced cycle ($p = 3\Delta_3$), the number of odd steps satisfies $K = \Delta_3 \le 91$.*

Proof. For $\Delta_3 \le 79$, we have $K \le 3\Delta_3 = p$ and $L = 2\Delta_3$. The condition $2^L > 3^K$ requires $K < L \cdot \log_2 3 \approx 1.585L = 3.17\Delta_3$. By explicit computation (verified in Lean 4 via `native_decide`): $2^{2\cdot 80-15} = 2^{145} < 3^{92}$, so $K \ge 92$ would require $L \ge 146$, i.e., $\Delta_3 \ge 73$. But $\Delta_3 \le 79$ forces $K \le 91$. □

THEOREM 5.7 (Cycle elimination for $\Delta_3 \le 79$). *No nontrivial Collatz cycle exists with $\Delta_3 \le 79$ odd steps per period.*

Proof. By Proposition 5.6, any such cycle has $K \le 91$. Hercher [6] has computationally verified that no nontrivial m-cycle exists for $m \le 91$ (extending earlier work of Simons and de Weger [11]). □

Remark 5.8 (The boundary at $\Delta_3 = 80$). The threshold $\Delta_3 = 79$ is optimal for Hercher's bound $m \le 91$. The continued fraction convergents of $\log_2 3$ determine which values of K are achievable: the convergents p_k/q_k are $(1,1), (2,1), (3,2), (8,5), (19,12), (65,41), (84,53), (485,306), \ldots$ At each odd convergent k, $2^{p_k} > 3^{q_k}$, so q_k is a candidate K-value. The seventh convergent has $q_7 = 306 > 91$, which lies beyond the Hercher threshold. For $\Delta_3 \ge 80$, the K-bound gives $K \le 3 \cdot 80 - 15 = 225$, which includes the convergent $q_7 = 306$ only conditionally.

Extending Hercher's bound to $m \le 200$ would push the threshold to $\Delta_3 \le 173$; to $m \le 306$ would cover all convergents up to q_7.

An alternative route uses Rhin's effective irrationality measure for $\log_2 3$ [10]: $|p/q - \log_2 3| > C/q^6$ for an effective constant $C > 0$. This gives a polynomial lower bound on $|2^L - 3^K|$ that, combined with the c_0 squeeze (5.2), could eliminate large cycles analytically. This approach is partially formalized in the Lean 4 development (`CycleElimination.lean`, `IrrationalityMeasure.lean`).

6. Main results

We now assemble the components into the main theorems.

Proof of Theorem A. Let $n \ge 1$ and assume the walk $w(t)$ has eventually linear drift with rate $\varepsilon > 0$.

By Theorem 4.4, the correction ratio $r(t)$ is eventually bounded by $B_r = 1/(1-\rho) + 1$.

By Corollary 4.5, the trajectory satisfies $a_t \le B = \lfloor B_r \rfloor + 2$ for all $t \ge T_1$.

By Corollary 4.6, the trajectory is eventually periodic.

By hypothesis, the only positive Collatz cycle is $\{1, 2, 4\}$. Since a_t is eventually periodic and must enter this cycle, we conclude $a_t = 1$ for some t. □

We now address the two hypotheses needed for the full conjecture.

Discussion of (H1). Hypothesis (H1) asserts that for every $n \geq 1$, the odd-step proportion $\nu_3(t)/t$ is eventually bounded below $p_{\rm eq} = 1/(1 + \log_2 3)$ by a uniform gap $\varepsilon > 0$, and that the linear bound $3\nu_3(t) \leq t + K$ holds for large t. In the formalization, the K-bound $3\nu_3(t) \leq t + K$ is the sole critical sorry (`finite_deficit_bound` in `DiophantineRepeller.lean`); the ε-gap follows since $p_{\rm eq} > 1/3$ (from $\log_2 3 < 2$, i.e., $3 < 4$). The converse also holds: if $T^k(n) = 1$ for some k, then the trajectory enters the cycle $1 \to 4 \to 2 \to 1$, which contributes exactly one odd step per three steps, so $3\nu_3(t) \leq t + K$ with $K = 3\nu_3(k) + 4$. This equivalence is formalized as `nu3_linear_bound_iff_reaches` in `Conclusion.lean`, confirming that the K-bound is the irreducible core of the Collatz conjecture.

The golden mean shift (Theorem 3.4) makes this plausible. The topological bound $1/\varphi^2$ already lies below $p_{\rm eq}$ by 0.005 (Remark 3.7). Computational evidence strongly supports the hypothesis: over 10^{10} starting values, the observed mean is $p_{\rm odd} = 0.3324$, well below $p_{\rm eq}$ (Section 8).

Density of obstructions. A natural strategy is to derive (H1) from Baker's theorem via the *structural pure-even sieve* (Definition 8.2, Section 8.4). At each torus scale $(\mathbb{Z}/3^k\mathbb{Z})^2$, certain cells are *structurally* pure-even: every trajectory visiting such a cell is forced to undergo at least two consecutive halvings, not by empirical observation, but by the arithmetic of the 2-adic valuation $v_2(3a_t+1)$. The Hensel lifting of the $3n+1$ map shows that these cells arise from congruence conditions on $a_t \bmod 2^m$ imposed by the multiplicative identity:

$$a_t \cdot 2^{\nu_2(t)} \equiv n \cdot 3^{\nu_3(t)} + C(t) \pmod{3^k}. \tag{6.1}$$

When ν_2/ν_3 lies in the Baker gap around $\log_2 3$ (i.e., $|\nu_2/\nu_3 - \log_2 3| < C'/(3^k)^{1+\varepsilon}$), these congruence conditions force $a_t \equiv 1 \pmod 4$, yielding $v_2(3a_t + 1) \geq 2$.

Baker's theorem ensures that the equilibrium line of slope $\log_2 3$ on the torus avoids all rational grid points, so the structural pure-even cells $\mathcal{S}_k$ occupy precisely the Baker gap—the region where the equilibrium line *would* pass if $\log_2 3$ were rational. A trajectory attempting to maintain ν_3/t near $p_{\rm eq}$ must follow the equilibrium line and therefore repeatedly cross $\mathcal{S}_k$-cells. Each crossing forces extra halvings, pushing ν_3/t below $p_{\rm eq}$. The Baker–Feldman bridge (Section 8.4) makes this argument precise.

What would suffice to prove (H1)? The remaining gap is a *visitation frequency* bound: one must show that every infinite Collatz trajectory visits $\mathcal{S}_k$-cells with frequency bounded below by a positive constant independent of n. An ergodic theorem for the Collatz transfer operator, or a spectral gap for the transition matrix on a finite torus quotient, would suffice [16]. Alternatively, one could show directly that the set of parity sequences sustaining $\nu_3/t > p_{\text{eq}} - \delta$ while avoiding $\mathcal{S}_k$ at all scales has Hausdorff dimension < 2. Tao's result [13] that almost all orbits attain almost bounded values provides partial evidence, though his methods are probabilistic and do not yield the uniform bound needed here.

Discussion of (H2) and (H3). Hypothesis (H2) is Hercher's computational verification [6] that no nontrivial m-cycle exists for $m \leq 91$, extending earlier work of Simons and de Weger [11]. This is established mathematics, treated as an external axiom in the formalization.

Hypothesis (H3) concerns cycles with $\Delta_3 \geq 80$. One route to eliminating these uses Baker's theorem [2] for the special case $\alpha_1 = 2$, $\alpha_2 = 3$:

$$|m \log 2 - n \log 3| > C / \max(|m|, |n|)^{1+\delta}$$

for effective constants $C, \delta > 0$. Combined with the c_0 squeeze (Corollary 5.5) and the Rhin irrationality measure [10], this could eliminate all large cycles analytically. The gap is not the mathematics but its formalization in Lean 4: the Gel'fond–Schneider proof chain underlying Baker's theorem has no Mathlib precedent (Section 7). Alternatively, extending Hercher's computation to $m \leq 306$ would suffice (Remark 5.8).

Proof of Theorem B. Assume (H1), (H2), and (H3).

Fix $n \geq 1$. By (H1), the walk $w(t)$ has eventually linear drift with some rate $\varepsilon > 0$.

By Theorem A, if the only positive Collatz cycle is $\{1, 2, 4\}$, then $T^k(n) = 1$ for some k.

It remains to verify the cycle hypothesis. For $\Delta_3 = 0$, a cycle is impossible (no odd steps means the trajectory decreases). For $\Delta_3 = 1$, this is Theorem 5.2: the only cycle is $\{1, 2, 4\}$. For $\Delta_3 \geq 2$ with $3\Delta_3 < p$, the strict inequality $a(t) \cdot 2^{\nu_2} < n \cdot 4^{\nu_3}$ contracts over the period. For $\Delta_3 \geq 2$ with $p = 3\Delta_3$ (the balanced case):

- $\Delta_3 \leq 79$: the K-bound (Proposition 5.6) gives $K \leq 91$, and (H2) (Hercher's theorem) eliminates all nontrivial m-cycles with $m \leq 91$ (Theorem 5.7).
- $\Delta_3 \geq 80$: excluded by (H3).

□

Remark 6.1 (Effective bounds). For $\varepsilon = 0.01$ (a modest drift rate): $\alpha = (\log_2 3 + 0.01)/(1 - 0.01) \approx 1.611$, $\rho = 3 \cdot 2^{-1.611} \approx 0.981$, and $B \approx 55$. For $\varepsilon = 0.001$: $B \approx 436$. The bound grows as $B \sim 1/\varepsilon$ for small ε.

7. Lean 4 formalization

The results of this paper have been formalized in Lean 4 using the Mathlib4 library. The formalization comprises approximately 7 700 lines across 32 files, building in 3 108 jobs.

7.1. *What is machine-checked.* The following results are fully proved (zero `sorry`):

Component	**File**	**Key theorem**
SFT library	`SFT.lean`	Shift-invariance, word avoidance
Golden mean shift	`FibCounting.lean`	Entropy $= \log\varphi$, $p_{\mathrm{odd}} = 1/\varphi^2$
Collatz–SFT bridge	`CollatzSFT.lean`	Parity sequence in golden mean shift
Branch locus	`BranchLocus.lean`	Cell trichotomy, pure-even forcing
Walk infrastructure	`Walk.lean`	Drift positivity, increment rules
Multiplicative independence	`Baker.lean`	$2^m = 3^n \Rightarrow m = n = 0$
Irrationality	`Baker.lean`	$\log_2 3 \notin \mathbb{Q}$
Correction ratio	`CorrectionRatio.lean`	Recurrence, universal inductive bound
Cycle elim. ($\Delta_3 \leq 1$)	`CorrectionRatio.lean`	$\Delta_3 = 0$: contradiction; $\Delta_3 = 1$: $\{1, 2, 4\}$
Cycle elim. ($\Delta_3 \leq 79$)	`SteinerCycle.lean`	Correction bounds + K-bound + Herch
Cycle linear form	`CycleElimination.lean`	Rhin lower bound, $\log(1+x) < x$ upper
Correction squeeze	`SteinerCycle.lean`	c_0 lower and upper bounds (proved)
Eventual periodicity	`CorrectionRatio.lean`	Pigeonhole principle
Syracuse framework	`Syracuse.lean`	Tao's odd-step iteration, identity
Syracuse–Drift bridge	`SyracuseDrift.lean`	$\nu_3 = k$, $\nu_2 =$ valSum at boundaries
Continued fractions	`ContinuedFraction.lean`	15 convergents of $\log_2 3$, power comparis
Irrationality measure	`IrrationalityMeasure.lean`	Rhin bound $\rightarrow$ cycle lower bound
Hensel attrition	`HenselAttrition.lean`	d consecutive $v_2 = 1 \Leftrightarrow x \equiv -1$ (mod 2^d
Diophantine repeller	`DiophantineRepeller.lean`	Baker cell separation, deficit tracking
Tunnel width	`TunnelWidth.lean`	Baker $\rightarrow$ Diophantine gap $\rightarrow$ pure-even
Deficit budget	`DeficitBudget.lean`	Deficit tracking, sliding window reductio
Weyl equidistribution	`WeylEquidistribution.lean`	Irrational rotation, lattice counting, safe
Cell error algebra	`SolenoidMixing.lean`	Cell error shift, Hensel–Baker conflict
Correlation decay	`CorrelationDecay.lean`	Autocorrelation vanishing, deficit compe
Skew product	`SkewProduct.lean`	Cocycle tracking, fiber coordinate identi
Borel–Cantelli	`BorelCantelli.lean`	Survivor density, danger density $\rightarrow 0$
Drift infrastructure	`Drift.lean`	K-bound $\rightarrow$ ε-gap, walk divergence
K-bound equivalence	`Conclusion.lean`	K-bound $\Leftrightarrow$ `collatzReaches`
Capstone	`Conclusion.lean`	`collatz_conjecture : CollatzConje`

7.2. *Remaining gaps: four axioms and the critical path.* Four external axiom declarations appear in the formalization. All four cite established results with published proofs; none represents genuinely open mathematics.

#	**File**	**Axiom**	**Status**
A1	`Baker.lean`	`baker_two_three`	Established (Baker 1966)
A2	`IrrationalityMeasure.lean`	`rhin_irrationality_measure`	Established (Rhin 1987)
A3	`SteinerCycle.lean`	`hercher_no_small_cycle`	Established (Hercher 2023)
A4	`WeylEquidistribution.lean`	`weyl_equidistribution`	Established (Weyl 1916)

Axiom A1 is Baker's theorem on linear forms in logarithms: $|m \log 2 + n \log 3| > C/\max(|m|, |n|)^\kappa$ for effective $C, \kappa > 0$ [1]. The Gel'fond–Schneider chain that would prove axiom A1 from scratch has two fully proved steps (Jensen zero counting via the identity principle; polynomial zero estimate via Vandermonde determinants) and three sub-lemma `sorry` declarations involving Blaschke products in `GrowthEstimates.lean`. Since axiom A1 is not on the critical path, these are vestigial.

Axiom A2 (Rhin's irrationality measure, $|p/q - \log_2 3| > C/q^6$) is an established result [10]. Axiom A3 (Hercher's verification that no nontrivial m-cycle exists for $m \le 91$) is a computational result [6]. Axiom A4 (Weyl's equidistribution theorem for irrational rotations [15]) is used in `WeylEquidistribution.lean`, where the lattice counting infrastructure is fully proved: for each row b, at most $\lceil 2\delta \rceil + 1$ cells satisfy $|a - \log_2 3 \cdot b| \le \delta$, giving a total dangerous cell bound of $N \cdot (\lceil 2\delta \rceil + 1)$ and safe cell density exceeding $1/2$ for large N.

The critical path. The sole critical-path `sorry` is `finite_deficit_bound` (`DiophantineRepeller.lean`): for every $n \ge 1$, there exists $D \in \mathbb{Z}$ such that $\mathrm{deficit}(n, t) \le D$ for all t. The proof chain is:

$$\texttt{finite\_deficit\_bound} \to \text{K-bound} \to \texttt{collatz\_conjecture}.$$

All arrows are sorry-free (`DiophantineRepeller.lean`, `CorrectionRatio.lean`, `Conclusion.lean`). The converse is also proved: `collatzReaches`(n) implies the K-bound (`nu3_linear_bound_iff_reaches` in `Conclusion.lean`), so bounded deficit is formally equivalent to the Collatz conjecture.

Note on the SlidingWindowCondition bug. An earlier version of the formalization used `SlidingWindowCondition` (SWC: $\forall t$, $\mathrm{deficit}(t + W) \le \mathrm{deficit}(t)$, i.e., deficit non-increasing over windows) as the critical-path target. This condition is *false* for many starting values: for $n = 27$, the deficit climbs from 0 to 12 during the transient before stabilizing in the $1 \to 4 \to 2 \to 1$

cycle. Computationally, $\sim$42.6% of starting values up to 10^7 violate SWC for all W. The corrected formulation (bounded deficit) avoids this error.

Structural support: the skew product decomposition. The Collatz map on the 2-adic integers $\mathbb{Z}_2$ has a natural *skew product structure* (`SkewProduct.lean`):

- *Base*: the 2-adic odometer $\sigma\colon \mathbb{Z}_2 \to \mathbb{Z}_2$ (adding 1), which is uniquely ergodic with respect to Haar measure.
- *Fiber*: the torus $\mathbb{T} = \mathbb{R}/\mathbb{Z}$.
- *Cocycle*: $\varphi(x) = \log_2 3$ if x is odd, 0 if x is even. The cocycle sum $\varphi_t = \log_2 3 \cdot \nu_3(t)$ equals the walk/fiber coordinate (proved sorry-free).

The mean cocycle value is $\log_2 3/2$, which is irrational (proved from `irrational_logb_two_three`). By Furstenberg's theorem on skew products over uniquely ergodic bases with irrational fiber rotation [5], the skew product is ergodic. Ergodicity gives $\nu_3(t)/t \to 1/3$ (Birkhoff averages), but the gap between "time averages converge" and "deficit is bounded" ($\exists D$, $\forall t$, $\operatorname{deficit}(t) \leq D$) remains open—this is exactly the content of `finite_deficit_bound`.

Three proved forces provide structural evidence that the deficit should be bounded:

(1) *Hensel attrition* (sorry-free, `HenselAttrition.lean`): a $v_2 = 1$ run of length d requires $2^{d+1} \mid (x+1)$, giving survival rate exactly 2^{-d}.
(2) *Baker cell separation* (sorry-free, reduces to A1): dangerous cells are Diophantine-separated with gap $> C/\max(|a|,|b|)^\kappa$.
(3) *Correlation decay* (sorry-free, `CorrelationDecay.lean`): for $d \geq 2$ and danger threshold $\delta < 1/2$, the autocorrelation of the danger indicator vanishes *exactly*: $C(d) = 0$. The shifted danger sets are proved disjoint on the torus, since the cell error shift $|d \cdot (1 - \log_2 3)| > 1$ exceeds twice the threshold.

Additional `sorry` declarations. Beyond the axioms, eight `sorry` declarations appear in the formalization. Three are equivalent to the Collatz conjecture (`nu3_linear_bound` in `Drift.lean`, `finite_deficit_bound` in `DiophantineRepeller.lean`, and `equidistribution_implies_deficit_bounded` in `WeylEquidistribution.lean`). One bridges Weyl equidistribution to the Collatz cell sequence (`nu3_linear_bound_from_weyl` in `WeylEquidistribution.lean`). One is off the critical path for cycle elimination (`steiner_cycle_large` in `SteinerCycle.lean`, handling $\Delta_3 \geq 80$; the $\Delta_3 \leq 79$ case is fully proved via Theorem 5.7). Three are Blaschke product sub-lemmas in `GrowthEstimates.lean`, vestigial since Baker's theorem is axiom A1.

7.3. *AI/LLM disclosure.* Per Annals policy, we disclose that Claude (Anthropic) was used as a coding assistant for the Lean 4 formalization and for drafting portions of the LaTeX manuscript. All mathematical content was developed and verified by the author. The Lean 4 proofs were checked by the Lean kernel, which is independent of any LLM.

8. **Computational evidence**

The following data are not formal proofs but provide quantitative support for the hypotheses. All computations were performed in C using the program `branch_locus.c`.

8.1. *The* 10*-billion run.* A complete run over all starting values $n \in [1, 10^{10}]$ was performed (19.5 hours wall time, 2.27×10^{12} total Collatz steps).

Quantity	**Value**	**Note**
Starting values N	10^{10}	
Total steps	2.27×10^{12}	
Mean p_{odd}	0.3324	Well below $p_{\text{eq}} \approx 0.387$
"11" violations	0	Over 2.27×10^{12} transitions
Transition fractions	ee = 33.45%	eo = 33.16%, oe = 33.39%

8.2. *Branch locus and tunnel structure.* At torus resolution $k = 144$, the branch locus (cells visited by both even and odd steps) and pure-even walls (cells visited only by even steps) were counted:

k	**Branch cells**	**Pure-even cells**	**Pure-odd cells**
144	16,371	659	0
729	16,577	893	0

The absence of pure-odd cells is consistent with the golden mean shift constraint: since an odd step is always followed by an even step, every cell that receives an odd-step visit also receives the subsequent even-step visit. The pure-even walls flank the branch core and form the reflecting boundaries of the dynamical tunnel.

8.3. *Dangerous cells and the Diophantine Repeller.* We define a cell (a, b) on $(\mathbb{Z}/3^k\mathbb{Z})^2$ to be *dangerous* if $E[v_2] < \log_2 3$ at that cell, meaning the mean 2-adic valuation of $3a_t + 1$ over all odd-step visits falls below the equilibrium threshold. If a trajectory could remain indefinitely in dangerous cells, its walk would have non-positive drift, defeating the argument of Theorem 4.4.

A large-scale two-pass computation (`v2_danger.c`) was performed over all $n \in [2, 10^{10}]$, yielding 7.55×10^{11} odd steps. In Pass 1, per-cell v_2 statistics were accumulated for torus scales 3^k, $k = 1, \ldots, 7$. In Pass 2, trajectories were replayed against the danger bitmap to measure consecutive runs of dangerous odd steps.

Box-counting: dangerous fraction by scale.

k	3^k	Occupied	Dangerous	Frac.	$E[v_2]$	Safe %
1	3	9	0	0	1.995	0
2	9	81	0	0	1.995	19.8
3	27	729	0	0	2.003	46.4
4	81	6,561	106	1.62%	2.074	71.4
5	243	16,253	709	4.36%	2.092	63.1

Scales $k \leq 3$ have zero dangerous cells; sufficient averaging at coarse scales pushes all cells above $\log_2 3$. At scale $3^4 = 81$, 106 cells (1.62%) are dangerous; at $3^5 = 243$, 709 cells (4.36%). The dangerous cells are precisely the continued fraction convergent neighborhoods of $\log_2 3$—the best rational approximants p/q to $\log_2 3$ modulo 3^k. The global mean is $E[v_2] = 1.995$, well above $\log_2 3 \approx 1.585$.

Residence time: the Baker Kick.

The crucial question is whether trajectories can sustain long consecutive runs in dangerous cells. The residence histogram gives a definitive negative answer.

Run length	**Count** (k=4)	**Count** (k=5)
1	2,627,762,566	2,927,121,626
2	81,596,040	96,928,973
3	597,843	781,804
4	137,676	173,948
5	4	8,869
6		566
7–12		61
Total	2,710,094,129	3,025,015,847
Max	5	12
Mean	1.031	1.033

Across 3.0×10^9 danger entries at scale 3^5, the maximum consecutive run is 12 odd steps, and the mean run length is 1.033. The run-length distribution decays exponentially: $P(\text{run} \geq \ell) \approx 0.033^{\ell}$. The tail fraction (runs ≥ 5) is 3×10^{-6}.

This exponential decay is the *Baker Kick*: trajectories entering a dangerous cell are expelled within 1–2 steps with overwhelming probability. The mechanism is Diophantine: the dangerous cells are isolated near the continued fraction approximants of $\log_2 3$, and Baker's theorem ensures these approximants are separated on the torus, preventing a trajectory from "surfing" along a chain of dangerous cells.

v_2 autocorrelation: mean reversion.

Lag	Autocorrelation	Pairs
1	-0.006	7.45×10^{11}
2	-0.082	7.35×10^{11}
3	-0.046	7.25×10^{11}
4	$+0.007$	7.15×10^{11}
5	-0.036	7.05×10^{11}

The v_2 autocorrelation is negative at all lags except lag 4. A low v_2 value (the dangerous case $v_2 = 1$) makes higher v_2 at subsequent odd steps more likely—mean reversion. The negative autocorrelation at lag 2 (-0.082) is particularly strong, suggesting two-step anti-persistence. This mean reversion provides a second mechanism (beyond the Diophantine isolation of dangerous cells) preventing sustained low-v_2 runs.

Summary. The v_2 danger analysis provides three lines of evidence for the Diophantine Repeller:

(1) *Isolation*: Dangerous cells ($E[v_2] < \log_2 3$) are confined to continued fraction convergent neighborhoods and occupy $< 5\%$ of visited cells at scale 3^5.
(2) *Expulsion*: The maximum consecutive residence in dangerous cells is 12 across 10^{10} trajectories, with mean 1.03—-trajectories are expelled almost immediately.
(3) *Mean reversion*: The v_2 autocorrelation is persistently negative, so a low v_2 step is self-correcting.

These observations support (H1): the walk cannot sustain non-positive drift, because the Diophantine structure of $\log_2 3$ prevents trajectories from lingering in the dangerous zone.

Hensel attrition: the 2-adic mechanism. The expulsion mechanism has an exact arithmetic foundation. A "hard dangerous" step is one with $v_2(3a_t +$

$1) = 1$, meaning $3a_t + 1 \equiv 2 \pmod 4$, i.e., $a_t \equiv 3 \pmod 4$. We prove (by induction on d):

Proposition 8.1 (Hensel attrition). *An odd number x can sustain d consecutive hard-dangerous steps ($v_2 = 1$ at each) if and only if $x \equiv -1 \pmod{2^{d+1}}$, i.e., the lowest $d+1$ bits of x are all* 1. *In particular, the fraction of odd residue classes permitting d consecutive hard-dangerous steps is exactly 2^{-d}.*

The proof proceeds by observing that at each hard-dangerous step, $x \mapsto (3x+1)/2$, and the condition $x \equiv -1 \pmod{2^{d+1}}$ maps to $(3x+1)/2 \equiv -1 \pmod{2^d}$: each step "peels off" exactly one trailing 1-bit from the binary representation. After a maximal hard-dangerous run of length d, the iterate has $x_d \equiv 1 \pmod 4$, forcing $v_2(3x_d + 1) \geq 2$ at the exit step—an arithmetic certainty, not a statistical tendency.

This gives two layers of danger attrition:

	Hard danger ($v_2 = 1$)	**Soft danger** ($E[v_2] < \log_2 3$)
Mechanism	Individual step valuation	Cell average on torus
Attrition rate	Exactly 2^{-d} (Hensel)	$\approx 0.06^d$ (empirical)
Provable	Yes (modular arithmetic)	Open

Hopping correlation: the stone-skipping phenomenon. To test whether trajectories can "skip" between isolated dangerous cells, we measured the step-to-step transition probabilities $P(D_{t+1} \mid D_t)$ and $P(D_{t+1} \mid S_t)$ for cells classified as dangerous at scale 3^k.

N	$P(D)$	$P(D \mid D)$	$P(D \mid S)$	Ratio
10^6	0.031	0.059	0.030	1.89
10^7	0.017	0.062	0.016	3.66
10^{10}	0.0042	0.032	0.0041	7.62

The hopping ratio $P(D \mid D)/P(D)$ exceeds 1 at every measured scale: trajectories in a dangerous cell are 2–8× more likely to hit another dangerous cell than baseline. This confirms that the dangerous cells form a *Diophantine network*—they cluster near the continued fraction convergents of $\log_2 3$, and the tripling map can occasionally map one convergent neighborhood to another.

However, the absolute conditional probability $P(D \mid D) \approx 0.032$ at $N = 10^{10}$ implies a 97% per-step escape rate. The rising ratio is an artifact of the shrinking denominator $P(D)$ as marginal cells average out above $\log_2 3$ with increasing N, while $P(D \mid D)$ stabilizes (decreasing from 0.06 to 0.032

as N grows from 10^6 to 10^{10}). The maximum consecutive dangerous run grows as $\sim \log N$ (observed: 5, 12 for $N = 10^{10}$ at $k = 4, 5$ respectively), consistent with exponential decay. The residence histogram at $k = 5$ shows $P(\text{run} \geq \ell) \approx 0.033^\ell$: 96.7% of danger entries have length 1, only 3.2% reach length 2, and runs of length ≥ 8 occur 12 times out of 3.0×10^9 danger entries.

Proof architecture. The Hensel attrition (Proposition 8.1) is formalizable in Lean 4 using only modular arithmetic; it does not require Baker's theorem. Combined with the Baker separation of dangerous cells (Section 8.4), this yields the proof chain (Section 7):

$$\text{Baker separation} + \text{Hensel attrition} \ \to\ [\text{bounded deficit: sorry}] \ \to\ \text{K-bound} \ \to\ \text{Collatz.}$$

The sole remaining gap is converting the density bound $P(\text{run} \geq d) = 2^{-d}$ (a statement about random starting values) into a per-trajectory bound (a statement about individual n). This requires showing that no trajectory can accumulate unbounded deficit—a question about the *mixing* properties of the Collatz map on the $(2, 3)$-solenoid, captured formally as the sorry `finite_deficit_bound`.

8.4. *Structural pure-even cells and the Baker–Feldman bridge.* The definition of pure-even cells used by the computational infrastructure (Section 8) is *empirical*: a cell is "pure-even" if no trajectory in a finite sample visited it with an odd step. This conflates an observation about specific trajectories with a property of the cell itself. We now give a *geometric* definition that depends only on the arithmetic of the cell, not on which trajectories happen to visit it.

Definition 8.2 (Structural pure-even cell). A cell $(a, b) \in (\mathbb{Z}/3^k\mathbb{Z})^2$ is *structurally pure-even* if every $n \geq 1$ that visits (a, b)—i.e., for which $(\nu_2(n, t) \bmod 3^k,\ \nu_3(n, t) \bmod 3^k) = (a, b)$ at some step t—has *even* iterate a_t at that step. In other words, every visit to the cell occurs at an even step (so the walk increment is $+1$), and no trajectory can take an odd step while occupying this cell.

Definition 8.3 (Forces double halving). A cell $(a, b) \in (\mathbb{Z}/3^k\mathbb{Z})^2$ *forces double halving* if every visit with an odd iterate a_t satisfies $v_2(3a_t + 1) \geq 2$, so that at least two consecutive halvings follow the odd step. Since structural pure-even cells have no odd visits, they force double halving vacuously.

The condition $v_2(3m+1) \geq 2$ is equivalent to $m \equiv 1 \pmod 4$: one checks $3 \cdot 1 + 1 = 4$, and in general $3m + 1 \equiv 4 \pmod 8$ when $m \equiv 1 \pmod 4$. This characterizes cells that *force double halving* (Definition 8.3): after an odd step at such an iterate, the trajectory divides by at least 4 before the next odd step, producing two even steps where only one was guaranteed. Structural pure-even cells (Definition 8.2) are a distinct, stronger property: the iterate is forced to be *even* by the torus arithmetic, so the step is even and no odd-step analysis applies.

Hensel lifting of the $3n+1$ map. The Collatz map lifts naturally to the 2-adic integers $\mathbb{Z}_2$. Given $n \in \mathbb{Z}_2$ with n odd, the map $n \mapsto (3n+1)/2^{v_2(3n+1)}$ is well-defined and continuous in the 2-adic topology. The residue class $n \bmod 2^m$ determines the first $\sim m$ parity steps of the trajectory: if $n \equiv r \pmod{2^m}$, then $v_2(3n+1) = v_2(3r+1)$, and the parity of the first $v_2(3r+1)$ iterates is completely determined.

At torus scale 3^k, the multiplicative identity (2.2) constrains $n \bmod 3^k$ and $n \bmod 2^m$ simultaneously. A cell (a,b) on $(\mathbb{Z}/3^k\mathbb{Z})^2$ forces $\nu_3 \equiv b \pmod{3^k}$, which via the identity imposes a congruence on $a_t \bmod 2^m$ for $m \asymp k \log_2 3$. When this forced residue class satisfies $a_t \equiv 0 \pmod 2$, the cell is structurally pure-even: every trajectory visiting it takes an even step, regardless of the starting value. When the forced class satisfies $a_t \equiv 1 \pmod 4$ (necessarily an odd iterate), the cell forces double halving (Definition 8.3).

The sieve. At each scale 3^k, the structurally pure-even cells form a subset $\mathcal{S}_k \subset (\mathbb{Z}/3^k\mathbb{Z})^2$. As k increases, the Chinese Remainder Theorem interleaves the 2-adic and 3-adic constraints, and the fraction $|\mathcal{S}_k|/3^{2k}$ stabilizes at a positive value. The cells in $\mathcal{S}_k$ act as a sieve: any trajectory entering one is forced to take an even step (walk increment $+1$).

LEMMA 8.4 (Forced even descent). *For each $k \geq 1$, let $\mathcal{S}_k$ denote the set of structurally pure-even cells in $(\mathbb{Z}/3^k\mathbb{Z})^2$ (Definition 8.2). Then:*

(i) *$\mathcal{S}_k \neq \emptyset$ for all $k \geq 1$.*
(ii) *For any $n \geq 1$ and any step t such that $(\nu_2(n,t) \bmod 3^k,\ \nu_3(n,t) \bmod 3^k) \in \mathcal{S}_k$, the iterate a_t is even, so the step is an even step and the walk increment is $+1$.*
(iii) *$|\mathcal{S}_k|/3^{2k} \geq c > 0$ for a constant c independent of k.*

Evidence. Part (i) is proved in Lean 4: cell $(0,1)$ is structurally pure-even at every scale 3^b (`tunnel_walls_positive_of_baker`). Part (ii) follows from the definition. Part (iii) is verified computationally for $k \leq 4$: at scale 3^2 (81 cells), 19 cells remain structurally pure-even after examining trajectories 1 through 10 over 50 steps; the count grows with k.

The Baker–Feldman bridge. The connection between Baker's theorem and the K-bound (`nu3_linear_bound`) proceeds in three steps.

(1) *Baker constrains the equilibrium line.* On the torus $(\mathbb{Z}/3^k\mathbb{Z})^2$, the equilibrium line has slope $\log_2 3$. Baker's lower bound

$$|m \log 2 - n \log 3| \;>\; C/\max(|m|,|n|)^{1+\delta}$$

implies that this line maintains distance $\gg (3^k)^{-\delta}$ from every rational grid point (p,q) with $0 \le p,q < 3^k$. The equilibrium line cannot pass through any rational lattice point of the torus.

(2) *Sieve cells and the Baker gap.* The structural pure-even cells $\mathcal{S}_k$ are located where the multiplicative identity (2.2) constrains the iterate to be even. Baker's theorem ensures these cells cluster near the equilibrium line but never on it, creating a "sieve belt" of forced even steps around the irrational slope $\log_2 3$.

(3) *The trajectory cannot ride the equilibrium.* Under equidistribution of the winding-number path on the torus, the trajectory visits $\mathcal{S}_k$-cells with positive frequency $\delta > 0$. Each such visit forces an even step (walk increment $+1$). Combined with the heuristic that the average 2-adic valuation $v_2(3a_t + 1)$ per odd step is $E[v_2] = 2 > \log_2 3 \approx 1.585$ (from near-uniform distribution of $a_t \bmod 2^m$), the walk has positive drift rate. The correction ratio bound (Theorem 4.4) then gives eventual boundedness, periodicity, and the K-bound via cycle elimination.

The rigorous version requires proving that the average v_2 per odd step exceeds $\log_2 3$ for every trajectory—not just under the random model.

Specifically, the following would suffice to close the gap:

(1) *Average halving bound*: For every $n \ge 1$, the average 2-adic valuation per odd step satisfies $\liminf_{k\to\infty} v_2(\tau_k)/k > \log_2 3$, where $\tau_1 < \tau_2 < \cdots$ are the odd-step times. Equivalently, $\nu_3(t)/t < p_{\mathrm{eq}}$ eventually. Under the random model, $E[v_2(3a+1)] = 2$ for odd a uniformly distributed, giving margin $2 - \log_2 3 \approx 0.415$.

(2) *Equidistribution*: A uniform distribution result for $a_t \bmod 2^m$ (conditional on a_t being odd) at a fixed depth $m \ge 2$. Even $m = 2$ suffices: if $a_t \bmod 4$ is uniformly distributed among $\{1,3\}$ at odd steps, then half have $v_2 = 1$ and half have $v_2 \ge 2$, giving $E[v_2] \ge 3/2$. Combined with the contribution from $v_2 \ge 3$ cases, this exceeds $\log_2 3$.

Tao's result that almost all orbits attain almost bounded values provides partial evidence, though his methods are probabilistic and do not yield the uniform bound needed here.

By the K-bound equivalence (`nu3_linear_bound_iff_reaches` in `Conclusion.lean`), the K-bound is formally equivalent to the Collatz conjecture for each n, so any route to the K-bound—whether through the Baker–Feldman bridge or otherwise—would resolve the conjecture.

Remark 8.5 (Computational results are not proofs). The computational data of Section 8 verify the golden mean shift constraint and the tunnel structure for $N = 10^{10}$, but they do not constitute formal proofs. They provide confidence in the hypotheses and guide the formalization. The observed $p_{\mathrm{odd}} = 0.3324$ is a mean over all trajectories; individual trajectories may temporarily exceed p_{eq}. The Baker–Feldman bridge above identifies the precise mathematical content needed to close the gap.

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